Escaping Inequality in India: The Role of English-Medium Instruction

Selected Excerpts

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What follows are excerpts from my 49-page independent research paper, "Escaping Inequality in India: The Role of English-Medium Instruction." The paper's key points are below.

1. **Question**: Does increasing baseline district-level EMI exposure (percentage of schoolgoing children enrolled in EMI) affect consumption growth over the next decade?
2. **Data sources**: 2007-08/2017-18 microdata and metadata from the National Sample Survey, 35 datasets from the 2001 Census, and 2019-20 GIS shapefiles.
3. **Identification**: District-level 2SLS (pop.-weighted linguistic distance from Hindi IV, state-clustered SEs). Individual-level probit of education selection, design-based SEs.
   - **Key clarification**: State FEs, essential for 2SLS exclusion restriction, trigger extreme multicollinearity. District matching & possible FD-2SLS redesign needed.
4. **Main result**: Current generated first-stage ($F = 0.8$) and second-stage estimates (0.1 pp, $p = 0.45$) are reported in the included tables. The preferred state-FE design is weakly identified, so the 2SLS point estimate is descriptive rather than credible causal evidence; weak-IV-robust diagnostics are retained with the replication outputs.
5. **Next steps**:
   - Repair 2SLS exclusion restriction with district harmonization changes, FD-2SLS.
   - Redesign models for spatial spillovers, migration, 'bad controls,' heterogeneity.
   - Make response variable robust to local variation in inflation and shocks.

The next ten pages include the following details:

6. **Research question**, contribution to literature, and summary/outline of paper. 10-11. **EMI exposure, probit model**, data wrangling, variable construction, missingness.
7. **Instrumental variable**, relevance, issues with exclusion restriction ($\kappa = 28730.84$).
8. **Interpretation of 2SLS**, key limits to interpretation, planned remedies. 26-27. **2SLS results**, both first- and second-stage.
9. **Interpretation of 2SLS**, small & noisy estimate, 'bad controls,' spillover attenuation. 32-33. **District harmonization** methods, assumptions. Some next steps (planned regressors, interactions, potential Bartik instrument).

Introduction and Literature Review

The Indian economy presents a series of paradoxes. The same economy whose GDP has grown at 6% annually on average for decades (International Labour Organization 2024, 2) has also seen stagnation if not losses in wages, employment, and labor force participation (see the corresponding figure in the full report), all while higher education has continued to develop an extremely strong, positive correlation with higher youth *unemployment*.[1] And though the International Labour Organization (ILO) traces this back to a labor supply which lacks the skills needed to fill available job openings (p. 174), high unemployment rates can be found for skilled laborers of all backgrounds, including those who received vocational education and training (Agrawal and Agrawal 2017).

[1]Note that lower levels of education are instead strongly correlated with youth *under*employment (International Labour Organization 2024, 130). There are no winners here.

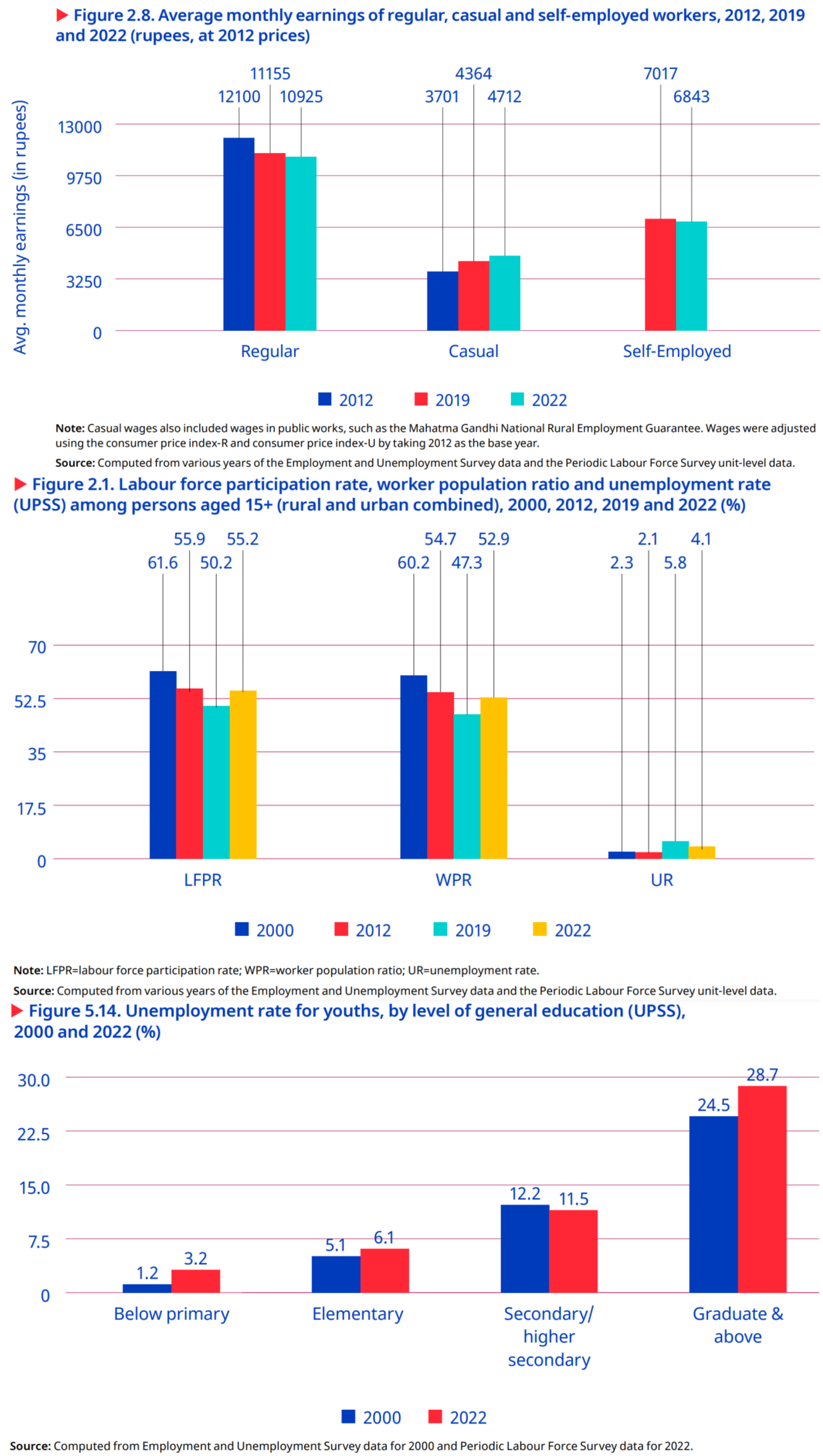

Figure 1: Trends in earnings, labor-force participation, and unemployment (ILO, 2024).

Appreciating this broader view helps reveal the degree to which unemployment, underemployment, and "jobless growth" (Graddol 2010) have, alongside wealth inequality, combined to undermine the degree to which many benefit from India's recent economic growth (Bharti et al. 2024). From here we adopt a bottom-up perspective—what can members of this "many" do to improve their lives in such a context?—and in doing so, our motivating question reveals itself to be two-fold: 1) If I am a household, what can I do to best ensure my children's future success in this context? And 2) if I am a local policy-maker, what can I do to best ensure my constituents' future welfare in this context?

The history of economics highlights *education* as one rich answer to these questions (Mincer 1974; Card 1999; Björklund and Kjellström 2002; Bhandari and Bordoloi 2006). The history of India, however, highlights another: *English*. In its capacity as global lingua franca (Melitz 2016), English is perceived as expanding mobility (Itani et al. 2015) and opening pathways to education at prestigious universities and jobs at wealthy multinational corporations across the world (Ramanathan 2014), particularly in the US (Bedi 2019, 32–33). But combine this with its role as domestic lingua franca (Clingingsmith 2014), and we see the number of intranational pathways English opens up as well: the mass off-shoring of millions of service jobs in the 1990s and the rise of information technology (IT) since have not only driven a rapid rise in India's GDP, but also a flurry of jobs whose higher socioeconomic returns (Chamarbagwala and Sharma 2011; Smokotin et al. 2014) are in large part only accessible to those who speak English (Mankiw and Swagel 2006; Shastry 2012).

Empirical study reveals the combined effect of these mechanisms. Using nationally-representative, household-level panel data, Azam et al. (2013) find that English fluency was, at least up until 2005, associated with a 34% hourly wage premium for men and a 22% premium for women; even for men and women with little English proficiency, the premia were still 13% and 10%. Refeque and Azad (2022) would later replicate this finding and extend it out to 2011-12. The strongest evidence for a causal effect of English skills on wages, however, comes from Chakraborty and Bakshi (2016), for whom West Bengal's 1983-2004 ban on English teaching in public primary schools constituted a natural experiment whose treatment group (those with increased "exposure" to the ban) faced a 26% decrease in future weekly wages relative to the control group (those less "exposed" to the ban)[2].

It is thus by combining our two answers—education and English—that we can begin to understand the demand for *English-medium instruction (EMI)*, or for the teaching of all school subjects in English, currently seen in India (Gooptu and Mukherjee 2023; Lahoti and Mukhopadhyay 2019; Chattopadhay and Roy 2017). "Language skills are human capital," after all (Chiswick and Miller 1995, 4), and "English-language skills [specifically] are a form of human capital" (Azam et al. 2013, 344); if the main mechanism by which education affects wages is human capital, then synthesizing education with English like this seems natural. Perhaps this is why EMI, as both a tool for social development (National Knowledge Commission 2009) and even as a tool for individual liberation,[3] is so often seen as a source of hope.

But the truth of EMI remains unclear and understudied. On one hand, there are limitations unique to EMI: instruction in one's mother tongue is associated with better academic participation and outcomes, both within India (Mohanty 2022; Nair 2015) and beyond (Bühmann and Trudell 2008), particularly when schooling quality and parents' socioeconomic status are controlled for.[4]

On the other hand, there are limitations unique to the many flaws of the Indian education system: low-effort (Singh 2013), even abusive (Centre for Budget and Policy Studies 2020, 218–22) teachers whose annual absenteeism rates have remained near 25% for decades (Muralidharan et al. 2017; Chaudhury et al. 2006, who also

[2]We will return to "exposure" again in the corresponding section of the full report

[3]Particularly from caste and the intergenerationally-rigid distributions of wage, employment, and occupation it enforces, as studied by Madheswaran and Singhari (2016) and Munshi and Rosenzweig (2006). See Mathur (2013) and Mathew and Srivastava (2011) for their advocation of, as well as Vij (2020) for their polemic for, EMI as a means of upwards growth.

[4]The key mechanism here appears to be the effect of exposure to a language at home on a child's ability to learn the language (Nair 2015; Erling et al. 2016). The dissimilarity between the languages spoken at home and the colonial languages chosen as lingua franca and media of instruction across the extremely linguistically diverse sub-Saharan Africa helps explain, per Laitin and Ramachandran (2022), the low rates of absolute learning and thus low rates of development in the region. Another potential mechanism, once hotly debated, is that childhood bilingualism generally confuses children, and that they most effectively learn with only one language while young (Hakuta 1986). The true effects of such bilingualism, however, seem to be much more nuanced; see Bialystok et al. (2012) for a review and for the effects of bilingualism into adulthood and old age.

find that only 45% of teachers at any one point in time were engaged in any kind of teaching-related activity)[5] coupled with inefficient resource allocation (Muralidharan and Sundararaman 2013; Gooptu and Mukherjee 2023) and poor commuting infrastructure (Kremer et al. 2005), all alongside schools which actively lie about their medium of instruction (Mathew and Srivastava 2011; Lahoti and Mukhopadhyay 2019), all of which is more commonly found in government schools than private. And yet the mismatch between expectation and reality can be so great that, even among the rural *private* schools surveyed by Lahoti and Mukhopadhyay (2019), "more than half (57%) of the children who [were] supposed to be [in EMI were] actually studying in the dominant regional language" (p. 55). Navigating such an environment requires parents to have enough information and social capital with which to identify higher-quality schooling/tutoring[6]—often the product of being well-educated themselves (Lahoti and Mukhopadhyay 2019; Blimpo et al. 2015)—but also to have a high enough income to afford this higher-quality[7] education (Muralidharan and Kremer 2008; Singh 2013).

The demand for both higher-quality education in general and EMI in particular has, since at least the 2000s, been increasingly met by a supply of 'low-cost'[8] private schools (James and Woodhead 2014; Srivastava 2013; Ramanathan 2016; Chattopadhay and Roy 2017).[9] Such schools lack the reputation typically afforded to higher-end private schools, meaning their students lack the social capital of other private school students (Ramanathan 2016). Conversely, more disadvantaged students are priced out of all schools except the free (if not negatively priced, see Muralidharan 2019) government schools. Note that these schools are not only associated with a lower quality of education (Muralidharan and Kremer 2008; Srivastava 2013; Muralidharan and Sundararaman 2015; Raju 2023), they are also associated with the most obvious signals of education quality, visible even to low-information parents, being worse-off e.g., worse facilities (Srivastava 2013; Central Square Foundation 2020, 54). Here we have evidence that school search ends with an inequality-amplifying positive assortative matching (Muralidharan 2019). Yet even amongst this new 'middle class' of schools, issues with EMI persist: the survey of Lahoti and Mukhopadhyay (2019) and the ethnographic study of Bhattacharya (2013) find children in low-cost private schools failing to learn both English as well as the subjects taught in it, further corroboration of the findings of Bühmann and Trudell (2008), Nair (2015), and Mohanty (2022). The reasons for this are likely two-fold: compared to parents who enroll their children in high-end private schools, those who select into low-cost private schools are typically 1) lacking in information and social capital, thus "[making] misguided evaluations of school quality based on visible factors" (Muralidharan and Sundararaman 2015, 1061–62); and 2) do not provide their children with as much community and out-of-classroom support for English (particularly when compared to "elite families" who often use English as a second language even within the family (Gupta 1997, 54–55)), preventing their oral skills from developing to the point where they can engage in class beyond rote memorization of the teacher's answers (Bhattacharya 2013; Treffers-Daller et al. 2022), to the extent such engagement is allowed (Brinkmann 2015).[10]

[5] The long-term consistency of this trend reflects the longevity of the civil service protections which produce it: limits on teacher hiring/firing power in government schools enable qualified teachers to extract such rents (Chaudhury et al. 2006). Even when rarely-absent teachers rationally advocate for such protections, they end up further deepening the culture of absence which Basu (2006) finds to explain the significant state-by-state heterogeneity in teacher absenteeism rates. The simple act of increasing the frequency of teacher monitoring, however, is enough to drastically reduce absenteeism (Kremer et al. 2005; Muralidharan et al. 2017).

[6] The best evidence for this comes from Andrabi et al. (2017) and Afridi et al. (2020), whose experimental treatments of education information provisioned to parents led to a more efficient market and improved student outcomes, particularly in the private schools nudged to improve quality by the treatment.

[7] Which in practice often equals private schooling; see Tooley et al. (2011).

[8] 'Low cost' may be a misnomer: poor households in the Delhi National Capital who sent their children to private school did so at the expense of 7-20% of their monthly incomes (Endow 2019).

[9] The demand for higher quality education almost always takes precedence over the demand for EMI, however. In 2014, for example, Joshi and Kumar (2017) show that 76% of households who chose private education did so for either a "better environment of learning" or to get a better education than a government school. Only 15.4%, however, listed EMI. Still, aside from private schooling, EMI seems to be the most frequently used proxy for high quality education

[10] It's worth noting our exclusion of the results from Muralidharan and Sundararaman (2015), despite it being a particularly noteworthy paper on the limitations of private schooling. By studying a randomized controlled trial of a private EMI voucher experiment, they discover that private schools did *not* lead to improvements in educational outcomes (although they did significantly lower cost). Tooley (2016), however, points out that this result may be driven by the fact that instructions for non-language subject exams were given in Telugu for government schools and English for private schools. He then finds "suggestive" evidence of what results would look like if all children took the same test (p. 580), and from there demonstrates a large, statistically significant advantage for private education.

Given all these mediators, nuances, and negations of its perceived effect, can EMI truly be seen as a source of hope? Should households turn to it to invest in their children's future? Does greater enrollment in EMI aid future development? Should policy-makers be promoting it as a tool for local development? To gain insight into these questions, we seek to answer the following one:

> What is the effect of exposure to English-medium instruction enrollment in 2007-08 on the growth of local consumption from 2007-08 to 2017-18?

First, note the improvements made here over the previous literature. The setting of 2007-2018 may be more relevant today than the settings of similar literature, all of which are closer to the advent of service job offshoring in the 1990s (e.g., Refeque and Azad 2022). Additionally, note that our response variable here is consumption, not wages. India holds a vast number of informal and self-employed individuals, meaning its labor force varies wildly in the degree to which wages reflect material well-being.[11]

Second, we must highlight that unlike much of this literature, we lack household panel data. We will thus follow previous papers (e.g., Martin et al. 2017) and aggregate data at the district level.[12] More specifically, we will take cross-sectional data from 2007-08 (National Sample Survey Office 2011) and 2017-18 (National Sample Survey Office 2022), aggregate each at the district level separately, and match districts across the time periods to construct a district pseudo-panel. District-level variables are developed in the corresponding section of the full report and the district matching method is explained in the corresponding section of the full report .

Our empirical methodology will then proceed in two parts. In the corresponding section of the full report, we study the composition of our key "EMI exposure" measure by estimating an individual-level probit of school participation in 2007-08. In the corresponding section of the full report, we move to the district level and implement two-stage least squares (2SLS): the weighted average of linguistic distance from Hindi in 2001 will be used to instrument for EMI exposure, onto whose fitted values we regress the change in log person-weighted real consumption.[13] The choice of instrument is discussed in the corresponding section of the full report.

Our 2SLS results will, at best, reflect the medium-run local general equilibrium effects of EMI exposure. On one hand, by aggregating away all individual or household-level variation, the ecological fallacy renders our 2SLS estimates uninterpretable at these lower levels. Additionally, school-going children in 2007-08 will still be quite young in 2017-18, and many may still be in school (particularly in more developed districts).[14]

In the preferred specification, the estimated effect of EMI exposure on person-weighted real consumption growth is statistically imprecise (the corresponding section of the full report), in line with previous literature on EMI and Indian education quality. Note, however, that our outcome includes households not directly exposed to EMI, meaning the effect we're studying would likely be inherently small regardless. Migration and spatial spillovers may also attenuate our results; evidence which both challenges and supports this is discussed in detail in the corresponding section of the full report).

[11] See Shrivastav and Husain (2026) and Bahl and Sharma (2024) for the frequency and wage effects of informality respectively. Wages may still be more responsive to EMI in the medium-run window we study, however. See the corresponding section of the full report for more.

[12] A district in India is akin to a county in the United States, but with an average population of 2 million. the corresponding table in the full report contains more information on district populations.

[13] Future work may incorporate both education selection and EMI exposure as endogenous regressors in a first-difference (FD) 2SLS design. The district-level aggregates of the corresponding table in the full report may be able to function as education supply-shifting instruments.

[14] Cohort-specific measures of EMI exposure may thus be incorporated into future drafts, as may robustness checks that use school-reported data on medium of instruction to construct EMI exposure—with the aforementioned caveat that schools may lie about having EMI (Mathew and Srivastava 2011; Lahoti and Mukhopadhyay 2019). Such surveys may also reveal whether EMI exposure is rigid over time. If so, this could allow us to expand the window of time over which we can interpret our results. Rigidity could also rule out the agglomeration of EMI discussed in the corresponding section of the full report as a potential attenuating force behind our results.

The Composition of Education Participation

Model and Variables

We are interested in how consumption growth is affected by EMI exposure (EMIE), defined over the full population of children ages 5-19:

$$\text{EMIE}_d = 100 \times \frac{\text{Num. children enrolled in EMI}}{\text{Num. children ages 5–19}} \text{ in district } d.$$

This all-child exposure combines the extensive enrollment margin with the choice of English-medium instruction conditional on enrollment. When medium is observed for every enrolled child, it equals the enrollment rate multiplied by the EMI share among enrolled children. To understand the first margin directly, we estimate the following probit model for whether child i is enrolled:

$$\begin{aligned}\Pr(\text{Enrolled}_i = 1 \mid X_i,\, Z_{d(i)}) = \Phi\Big(&\beta_0 + \beta_1\,\text{Age}_i + \beta_2\,\text{Female}_i + \beta_3\,\text{HHSize}_i + \beta_4\,\text{Urban}_i \\ &+ \sum_{r\in R}\gamma_r\,\text{Religion}_{ir} + \sum_{s\in S}\delta_s\,\text{SocialGroup}_{is} + \sum_{d\in D}\theta_d\,\text{SchoolDist}_{id} \\ &+ \sum_{f\in F}\psi_f, \text{FatherEduc}_{if} + \sum_{u\in U}\eta_u\,\overline{u}_{d(i)} + \phi\,\overline{\text{EnrollmentCost}}_{d(i)}\Big) \qquad (1)\end{aligned}$$

where

$$\begin{aligned} R &= \{\text{Muslim, Christian, Sikh, Jain, Buddhist, Zoroastrian, None}\},\\ S &= \{\text{Scheduled Tribe, Scheduled Caste, OBC}\},\\ D &= \{\text{1–2km, 2–3km, 3–5km}, \geq \text{5km}\},\\ F &= \{\text{Illiterate; Literate, no school; Literate, school} < \text{primary;}\\ &\qquad \text{Primary, Upper primary, Secondary, Higher secondary, Postsecondary+}\},\\ U &= \{\text{Educ. free available, Tuition waived, Scholarship/Stipend,}\\ &\qquad \text{Textbooks received, Stationery received, Mid-day meal, etc.}\}\end{aligned}$$

and

$$\begin{aligned}\overline{u}_{d(i)} &= \text{Average value of } u \in U \text{ in district } d \text{ amongst enrolled children,}\\ \overline{\text{EnrollmentCost}}_{d(i)} &= \text{Average value of EnrollmentCost in } d \text{ amongst enrolled children.}\end{aligned}$$

There are two kinds of regressors here: those in X_i, the variables unique to child i, and $Z_{d(i)} = \{\overline{u}_{d(i)} : u \in U\} \cup \{\overline{\text{EnrollmentCost}}_{d(i)}\}$, the variables which have been aggregated at the level of the district $d(i)$ where child i lives. Without this aggregation, neither u nor EnrollmentCost, both endogenous to enrollment, would be well-defined for children not in school. Our aggregation can thus be understood as a way of handling missingness; whereas EnrollmentCost_i was missing for 120,062 children (as reflected in the corresponding table in the full report), $\overline{\text{EnrollmentCost}}_{d(i)}$ was missing for 7,109. Observations still missing a relevant variable were then dropped.[15]

The district-level averages can be interpreted as reflections of local supply-side factors, namely the general availability or generosity of educational inputs in a child's local environment. Their marginal effects (see the

[15]Three such variables had missing values before dropping: the district-average $\overline{\text{EnrollmentCost}}_{d(i)}$ (7109 missing values), the distance from the nearest primary class (312), and the father's education proxy (67). We find that none of these variables are MCAR (missing completely at random) under logistic regressions with missingness indiciators and adjusted p-values (per Benjamini and Hochberg 1995). The lowest McFadden pseudo R^2 among the three was a 0.246 on father's education, which had three predictors significant at a level of 0.05 (that is, three variables in the data could predict its missingness with adjusted p-values below the 0.05 threshold). The highest pseudo R^2 was a 0.31 for $\overline{\text{EnrollmentCost}}_{d(i)}$, which had nine significant predictors. Note that prior to this, many other NAs in $\overline{\text{EnrollmentCost}}_{d(i)}$ were handled using codebook-informed aggregations and redefinitions of the variables summed to derive EnrollmentCost_i.

corresponding table in the full report) may reflect how changes in these district-level characteristics affect the probability a child enrolls in education—or in other words, the elasticity of enrollment with respect to local schooling conditions.[16]

Following Azam et al. (2013), Munshi and Rosenzweig (2006), and Singh (2013), we use father's education to proxy a child's ability before schooling. We use the father despite evidence indicating that the mother's education is more strongly associated with early life human capital development.[17] Taking the average of the mother's and father's education is not possible as education level is recorded by the National Sample Survey Office (2011) as a categorical variable. Future work may attempt to create an index variable for this.

The National Sample Survey Office (2011) do not explicitly identify parent-child relationships. Instead, they only provide each person's relationship to the head of their household. Father's education is thus proxied by the education of the head of the household if they are male (average age here is 14); else as the married child of the head if they are male (married men often live with their elderly parents; average age of 18.1); else as the parent/parent-in-law of the head if they are male (average age of 13.5). We do not include male spouses of the head, whose average age is 11.8 and share of children (aged <18) is 0.896 in our data.[18]

Summary statistics for numeric variables are given in the corresponding table in the full report, and for categorical variables in the corresponding table in the full report.

[16] Future work will likely use these district-level averages as instruments for enrollment, an endogenous variable to be included alongside EMIE in a first-difference (FD) 2SLS framework. It may be possible to use this probit as a control function or in a two-stage residual inclusion (2SRI) estimation.

[17] Card (1999) shows that in the US, among people born from before 1920 to 1964, mother's education better predicts school completion among Black people, whereas father's education better predicts for cohorts born 1955-1964. In general, however, he finds mother's education to be the the slightly better predictor of school completion. It's interesting to note that this is a quantitative outcome: one might also expect mother's education to better predict the *quality* of early childhood investment too, with father's education relating more to the quantity of early childhood investment.

[18] This is a clear indication of severe, systematic misclassification error. Further checks on the presence of misclassification imply that the codes for unmarried children and spouses of heads were simply flipped; the average age of male "unmarried children," for example, is 18.1 in the data. Future work will either incorporate the mislabeled "unmarried children" into this proxy chain for father's education, or use the household head's education alone as proxy.

Table 1: Summary Statistics for Enrollment Participation Model (Numeric Variables)

Variable	Min	1Q	Med	3Q	Max	Mean	SD	N
Age	5.00	8.00	12.00	16.00	19.00	11.89	4.22	127181
Household size	1.00	4.00	5.00	7.00	30.00	5.91	2.40	127181
Enrollment cost (Rs.)	0.00	2,315.00	4,900.00	8,750.00	338,300.00	7,050.07	9,285.69	7181
District-level aggregates:								
Educ. free available? (Yes = 1)	0.00	0.48	0.69	0.85	1.00	0.65	0.23	127181
Tuition waived?	0.00	0.00	0.01	0.02	0.48	0.02	0.04	127181
Scholarship/Stipend?	0.00	0.01	0.05	0.20	0.90	0.13	0.17	127181
Textbooks received?	0.00	0.37	0.52	0.66	0.97	0.51	0.20	127181
Stationery received?	0.00	0.01	0.04	0.11	0.95	0.08	0.12	127181
Mid-day meal or more received?	0.02	0.33	0.42	0.55	0.90	0.43	0.17	127181
Avg. district enrollment cost	84.55	3060.61	5158.42	7661.91	58055.64	6010.16	4352.63	120072

Note: Min. = minimum; 1Q = first quartile; Med. = median; 3Q = third quartile; Max. = maximum; Mean = arithmetic mean; SD = standard deviation; N = number of observations.

Table 2: Summary Statistics for Enrollment Participation Model (Categorical Variables)

Variable	Values	Mode	Pct. Mode	Least Freq.	Pct. Least Freq.	N
Sex	Male, Female	Female	53.4	Male	46.6	127181
Religion	Hindu, Muslim, Christian, Sikh, Jain, Buddhist, Zoroastrian, Other	Hindu	75.2	Other	0	127181
Social group	Other, Scheduled Tribe, Scheduled Caste, Other Backward Class	Other Backward Class	39.5	Scheduled Tribe	14.2	127181
Urban	Rural, Urban	Rural	67.5	Urban	32.5	127181
Distance of nearest primary class	d<1km, 1km <= d <2kms, 2kms<= d <3kms, 3kms <= d <5kms, d>=5kms	d<1km	91.7	3kms <= d <5kms	0.1	126869
Father's education	Illiterate, Literate, no school, Literate, school < primary, Primary, Upper primary, Secondary, Higher secondary, Postsecondary+	Illiterate	55.8	Literate, no school	1.4	121978

Note: Values = all possible values; Mode = most frequent value; Pct. Mode = percent of observations taking the modal value; Least Freq. = least frequent value; Pct. Least Freq. = percent of observations taking the least frequent value; N = number of observations.

Results and Discussion

Average marginal effects for numeric variables and counterfactual comparisons (relative to the reference level) for categorical variables (both referred to as AME henceforth) are reported in the corresponding table in the full report, which has been derived using the work of Arel-Bundock et al. (2024) In line with advice from Nowosad and Tennekes (2021), we report the AME instead of marginal effects at the mean.

the corresponding table in the full report has been calculated over all observations, with standard errors clustered at the rural village and urban block level (no district clustering) derived using the delta method and automatic differentiation.[19] All reported values are probabilities: that AME_{Age} = -0.028 in the table below thus means that, on average, a one-year increase in age is associated with a 2.834 percentage point decrease in the probability of being enrolled by, holding all else constant (Hansen 2022). The counterfactual comparison AME_{Muslim} = -0.126, meanwhile, where 'Hindu' is the reference category to 'Muslim', means that the average change in the probability of a Hindu child being enrolled if they were hypothetically made Muslim is -0.126, keeping all else fixed at observed values.[20]

Table 3: Average Marginal Effects and Counterfactual Comparisons for Enrollment Probit

	Enrolled (1 = yes)
	(1)
Age (years)	-0.028***
	(0.000)
Female (ref: Male)	0.044***
	(0.003)
Household size	-0.004***
	(0.001)
Religion: Muslim (ref: Hindu)	-0.126***
	(0.007)
Religion: Christian	-0.043**
	(0.014)
Religion: Sikh	0.040
	(0.025)
Religion: Jain	-0.001
	(0.012)
Religion: Buddhist	0.047
	(0.024)
Religion: Zoroastrian	0.101**
	(0.034)
Religion: Other	0.252***
	(0.003)
Social group: Scheduled Tribe (ref: Other)	-0.102***
	(0.008)
Social group: Scheduled Caste	-0.075***

[19] Dawood and Megahed (2023) provide a good treatment of automatic differentiation, describing it as a process which is "neither numeric nor symbolic, nor is it a combination of both…. In contrast to the more traditional numerical methods based on finite differences, auto-differentiation is 'in theory' exact, and in comparison to symbolic algorithms, it is computationally inexpensive." The `marginaleffects` package itself claims automatic differentiation is both faster and more accurate for some models (Arel-Bundock et al. 2024).

[20] Such interpretations are possible because AME is averaged across all observations, making them more generalizable than metrics like marginal effects at the mean (Nguyen 2020). But their interpretability is still also dependent on a linearity assumption which may not be valid. Future work may thus abide by the argument of Scholbeck et al. (2024), that no one single metric can be used to summarize a non-linear prediction function's feature effects, and may instead report conditional feature effect estimates by applying their concept of forward marginal effects within population subgroups.

Table 3: Average Marginal Effects and Counterfactual Comparisons for Enrollment Probit *(continued)*

	(1)
	(0.006)
Social group: Other Backward Class	-0.029***
	(0.005)
Urban (ref: Rural)	-0.003
	(0.005)
Distance 1–2km (ref: <1km)	-0.077**
	(0.026)
Distance 2–3km	-0.001
	(0.010)
Distance 3–5km	0.004
	(0.066)
Distance > 5km	0.036
	(0.035)
Father's educ.: Literate, no school (ref: Illiterate)	0.097***
	(0.017)
Father's educ.: Literate, school < primary	0.127***
	(0.006)
Father's educ.: Primary	0.142***
	(0.005)
Father's educ.: Upper primary	0.195***
	(0.006)
Father's educ.: Secondary	0.212***
	(0.007)
Father's educ.: Higher secondary	0.247***
	(0.008)
Father's educ.: Postsecondary+	0.245***
	(0.009)
Educ. free available (ref: No)	-0.089***
	(0.015)
Tuition waiver received	-0.083
	(0.057)
Scholarship/Stipend received	0.030*
	(0.012)
Textbook(s) received	0.067***
	(0.018)
Stationery received	0.013
	(0.020)
Mid-day meal, etc. received	-0.011
	(0.019)
Enrollment cost (Rs.)	0.000
	(0.000)
Observations	127181

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$
NSS 64th round; design-based SEs in parentheses.

The goal of this paper is to study how well household optimization (through education participation and EMI enrollment) and local policy changes (aimed at shifting education supply or EMI enrollment) can lead to upward mobility given the context they exist in: one of inequality and jobless growth. In our results we do indeed find supply and social structure to be meaningful predictors of how households optimize via education enrollment.

We first look at household-level determinants of who gets to stay in school. Our data is of children aged 5-19, so as a 5-18 year old child grows one year older, the probability they stay enrolled in school decreases by -2.834 percentage points on average, indicative of either economic necessity, the opportunity cost of school, or even disillusionment with school quality growing in age.[21] We also see that each additional household member lowers enrollment probability by -0.368 percent. A negative value is expected—household resources should dilute as size increases—but the slight value and high significance are notable given the quantity-quality tradeoff of e.g., Becker (1969), plus empirical tests of it in India. Kumar and Kugler (2011) find that, by using children's educational attainment as a proxy for quality, cross-sectional dataset from 2007-08 indicates the tradeoff is heterogeneous in India, stronger given rural areas, low-caste children, illiterate mothers, and poor children. Conversely, by instrumenting family size Onur and Velamuri (2021) remove the quantity-quality tradeoff entirely from their panel data, whether quality is proxied by educational expenditures on a child or test scores. Enrollment is such a relatively low-cost, extensive-margin measure for quality, however, that it makes sense that neither the heterogeneity nor non-existence of these previous studies shows up here. The significance of our coefficient is understandable given the many correlates of fertility and schooling amongst our regressors, whereas the small magnitude of our estimate may arise from a non-linearity in enrollment (e.g., if household size only begins to constrain at a high enough level), from economies of scale in enrollment costs, and from the aforementioned low quality threshold that is enrollment. This last factor also colors our interpretation of the relatively high probability of female enrollment (4.417 percent more than male enrollment): while it likely reflects the success of programs such as the Sarva Shiksha Abhiyan and the Kasturba Gandhi Balika Vidyalayas girls' schools (Press Information Bureau 2008), it also tells us nothing about the quality of education beyond enrollment.

While both Muslims and Christians are less likely to be enrolled than Hindus (whereas Buddhists and Zoroastrians are more likely), the sharp Muslim penalty of -12.604 percent echoes Borooah and Sabharwal (2021), who find that even upper-class Muslims enroll at lower rates. Even greater statistical significance can be found among Scheduled Tribe, Scheduled Caste, and Other Backward Class members, all of whom have negative coefficients, with Scheduled Tribes faring the worst at -10.217 percent. While we should be suspicious of causal interpretations, it is likely these coefficients reflect the way structural factors i.e., those which cannot be changed through individual actions alone, constrain the choice set over which households optimize over. For example, if we use chronic teacher absenteeism to reflect school quality, then we can use Kremer et al. (2005) to see that better infrastructure and proximity to a paved road correlate with increased school quality, as does more frequent school monitoring per both Kremer et al. and Muralidharan et al. (2017). If we assume that the physical and political infrastructure needed to do this is decreasing in the share of ST/SC/OBC members in the area, then this only decreases the opportunity cost of labor for children in these groups.

Looking deeper into the supply side of education, it seems that all variation which could potentially be explained by enrollment cost has, at best, been consumed by other variables, implying direct costs matter less than social barriers and other correlates of supply-side factors. While one would expect the causal effect of an exogenous shock in scholarships/stipends, stationery, or textbooks to match the sign of their positive coefficient estimates, only textbooks show up as statistically significant. Instead, our other significant result is the -8.856 percent change in enrollment associated with a marginal change in the share of students within one's district who attend a school which charges no tuition fees (i.e., where 'Educ. freely available'). National Sample Survey Office (2011) documentation indicates that such schools include most if not all government schools, as well as private schools in some states up to a certain level of education. Building off observations in the corresponding section of the full report that government schools tend to have worse facilities, chronic

[21] In 2002, the right to free and compulsory education for all children aged 6-14 was enshrined by Jain (2002) into the Constitution of India. One might expect this to explain our AME_{Age} result as the product of a steep drop in enrollment between the ages of 14 and 15 which violates the linearity assumption underlying the interpretation of AME values (Scholbeck et al. 2024). But the amendment neither enforced nor implemented, instead only mandating future legislation to do so; such legislation was not passed until 2009 Department of Higher Education (2010). Something which did exist in 2007-08, however, was the Sarva Shiksha Abhiyan program, which had been allocating funds for school construction, facilities management, teacher training, girls' education, and so on since 2001-02 (Press Information Bureau 2008). One would expect this to increase (as in make more positive) our AME_{Age} = -2.834 estimate, and it likely has increased our $\text{AME}_{\text{Female}}$ = 4.417 estimate.

teacher absenteeism, and so on, this coefficient seems to imply that when the modal inside option is poor-quality schooling, the marginal child is more likely to have taken the outside option not be enrolled. If this interpretation is correct, then comparing the variables' s-values[22] using Mansournia et al. (2022) reveals just how much evidence we find for government schools being a negatively-priced deterrent (a la Muralidharan 2019): while both 'Textbook(s) received' and 'Educ. free available' have similar effect sizes and standard errors, the former has an s-value of 12.6, meaning that the data provided 12.6 bits of information against the null hypothesis (a coefficient of zero), whereas the 'Educ. free available' result communicated a far higher 27.5 bits against its null.

Finally we have the most economically and statistically significant terms: the father's (or father proxy's) education level. We can interpret this in multiple ways: as a proxy for financial ability (via the relationship between a father's education and the household's permanent income), for heritable cognitive ability, for a child's valuation of education (via intergenerational transmission of tastes and values), and even for access to educational and labor market information (via social capital). It's possible that these variables, insofar as they correlate with fertility and schooling choices, absorbed some of the variation which otherwise would've been explained by, say, household size. But the size and significance of these coefficients may imply something further: that a household's financial/cognitive ability, tastes/preferences, and social capital/information networks affect both household optimization and the choice set available for them to optimize over.

Instrumental Variable

Our instrument for EMIE_d is LingDistance_d, the speaker-weighted mean Shastry distance from Hindi among mapped mother tongues with positive distance in district d in 2001. The mechanism behind this may seem odd: why choose linguistic distance from Hindi as opposed to, say, English? India's large linguistic diversity[23] predisposes it to have large communication costs, large coordination costs, and thus hindered economic development within its borders (Nakagawa and Sugasawa 2021). To decrease such frictions in the market and in public goods provisioning (Liu and Pizzi 2016), both Hindi and English have been thrust into the roles of competing domestic lingua franca (Clingingsmith 2014), meaning Hindi- and English-medium schools can be found across the country (Shastry 2012, 294).

As the 'distance' between one's mother tongue and Hindi increases, we expect the opportunity cost of learning English over Hindi to decrease. Because of this, we also expect individuals with a mother tongue further from Hindi to be more likely to enroll in EMI. "That distance from Hindi induces people to learn English" and even "schools to teach English" is in fact precisely what Shastry (2012, 289) claims. If we can then show that local linguistic distance induces exogenous variation in local EMI, we can wield it as an instrumental variable (IV) to isolate EMIE_d's effect on consumption growth. Shastry helps us once more here, evidencing both the relevance (pp. 299-301) and exclusion restriction (pp. 302-305) of LingDistance_d's effect on EMIE_d.

We are currently unable to replicate her justification of the exclusion restriction, however. Her argument centers on a map depicting the geographical balance of her residual variation, made possible despite the distinct regional divide in the corresponding figure in the full report thanks to state fixed effects. In our case, the state-FE specification has condition number $\kappa = 24.92142$ even though the production pseudo-panel uses unique Census-2001 target districts and passes the lineage validation gates. The remaining instability is therefore an identification problem rather than an artifact of duplicated panel rows: linguistic distance is strongly structured across states, leaving little conditional variation once state fixed effects and predetermined district controls are included.

[22] Given a p-value, we can define the s-value as $s = -\log_2(p)$ (Mansournia et al. 2022).
[23] Perhaps the largest in the world, per one metric from Gurevich et al. (2025).

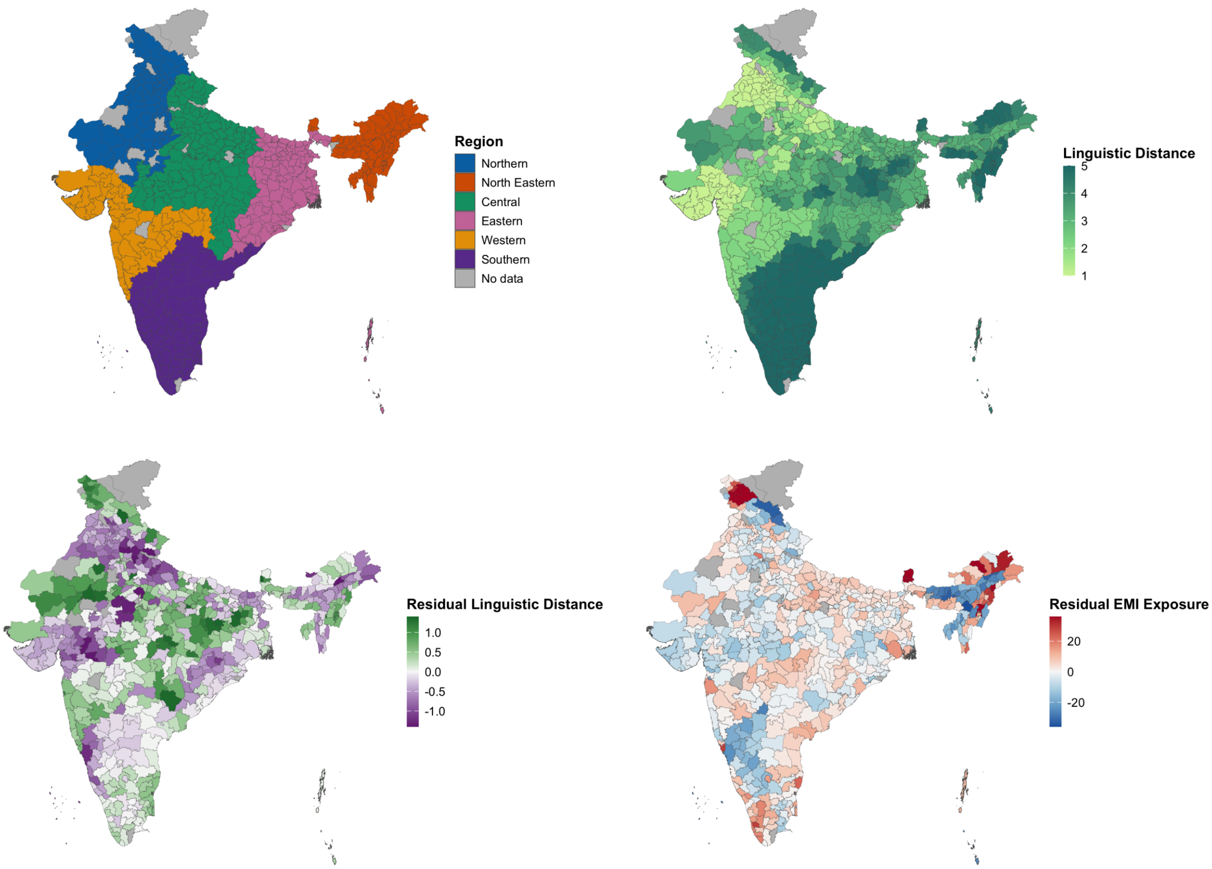

Figure 2: From left to right: regions of India and linguistic distance from Hindi. District-level data, from the 2001 Census of India.

Part of this absorption is structural: the States Reorganization Act of 1956 aligned many state lines with linguistic lines, a precedent later states have often followed (Sarma 2025). This is why the state-FE first stage is an important stress test rather than a nuisance to be removed. Future work should seek excluded variation that survives within-state conditioning, potentially through time-varying supply or demand shocks and weak-IV-robust estimators rather than dimensionality-reduction methods that change the identifying variation.

In the meantime, we rely on heuristic justifications of our instrument. Its relevance is plausible: districts with higher linguistic distance from Hindi should have higher EMI demand due to the relatively lower opportunity cost of learning English. They should also have relatively lower cross-language coordination costs for setting up EMI as opposed to Hindi-medium instruction, allowing EMI supply to rise more. Shastry justifies this at the state level; future work must replicate this at the district level.

The exclusion restriction, meanwhile, requires us to assume that after conditioning on our controls, a district's linguistic distance is a) predetermined and b) plausibly orthogonal to all later shocks and trends in consumption except via education. We could demonstrate (b) by replicating Shastry's visualization of geographically balanced residuals, though this will require state/region fixed effects (and possibly spatial correlation robust SEs; see the corresponding section of the full report). More work will be needed to replicate her justification of (a): not only does she show that district-level linguistic distance in both 1991 and 1961 are strongly correlated, she also shows that they both increase the within-state share of a language's native speakers who learn English in 2001, as well as the in-state level of EMI supply in 2001. That linguistic distance remains consistent over decades implies little inter-district migration, in line with most literature in the corresponding section of the full report.[24] Our exclusion restriction may still fail, however, if districts further from Hindi benefit differently from pre-existing human capital, access to non-Hindi media, trade relations, NGO/government funding, and so on compared to districts closer to Hindi. Further work will be needed to address this.

LingDistance_d uses Shastry's simplest and most frequently used measure: a 0-5 integer scale reflecting the "degrees" of separation between one's mother tongue and Hindi, developed with Harvard linguist Jay Jasanoff.[25] We assign those published distance categories through the tracked Census-2001 language concordance, then take the speaker-weighted mean among mapped languages with positive distance in each district. Coverage and alternative constructions, including the older top-three-language mean and the five nonzero distance shares, are reported in extended diagnostics rather than substituted into the public specification.

The Effect of EMIE: 2SLS Estimation and Main Results

Model and Variables

Inspired by the measures of public school exposure in Chakraborty and Bakshi (2016), we define district EMI exposure (EMIE) as the survey-weighted share of children ages 5–19 who are both enrolled and observed in English-medium instruction:

$$\text{EMIE}_d = 100 \times \frac{\text{Num. children ages 5–19 enrolled in EMI}}{\text{Num. children ages 5–19}} \text{ in district } d.$$

This all-child exposure combines the extensive margin of school enrollment with the medium-of-instruction choice among enrolled children. We are interested in how variation in this exposure affects district real consumption growth from 2007-08 to 2017-18. Our preferred outcome is the log difference in person-weighted real monthly consumption. Nominal household expenditures are first deflated using state-sector temporal indices and Tendulkar spatial relatives, then aggregated to districts with household survey weights and household size. We use LingDistance_d as an instrument for EMIE_d (see the corresponding section of the full report). The public 2SLS specification is

$$\begin{aligned} \text{EMIE}_d &= \gamma_0 + \gamma_1 \,\text{LingDistance}_d + X_d'\gamma + \alpha_s + v_d, \\ \Delta \log C_d^{real} &= \beta_0 + \beta_1 \,\widehat{\text{EMIE}}_d + X_d'\beta + \alpha_s + u_d, \end{aligned} \tag{2}$$

where X_d contains predetermined Census 2001 controls and α_s denotes Census-2001 state fixed effects. Summary statistics for all variables in this model, including the controls k in the vector X_{kd}, are provided in the corresponding table in the full report. Maps of these variables, created using 2001 boundaries, are presented in the corresponding figure in the full report and the corresponding figure in the full report.

[24] Though perhaps not in line with Economic Division (2017), whose gravity model for migration purportedly shows that while political barriers constrain migration, linguistic ones do not: "not having a common language does not impede the flow of migrants" (p. 275). Their 'common language' proxy, however, only indicates if a state is primarily Hindi-speaking or not. If Hindi truly does compete with English as national lingua franca, as Clingingsmith (2014) claims, then insignificance is expected.

[25] Shastry (2012) finds similar results across different measures, including the share of speakers with mother tongues three or more degrees away. One such measure relies on currently inaccessible data on the percent of cognates shared with Hindi. Anderson et al. (2025) may enable us to construct a cognate-based robustness measure.

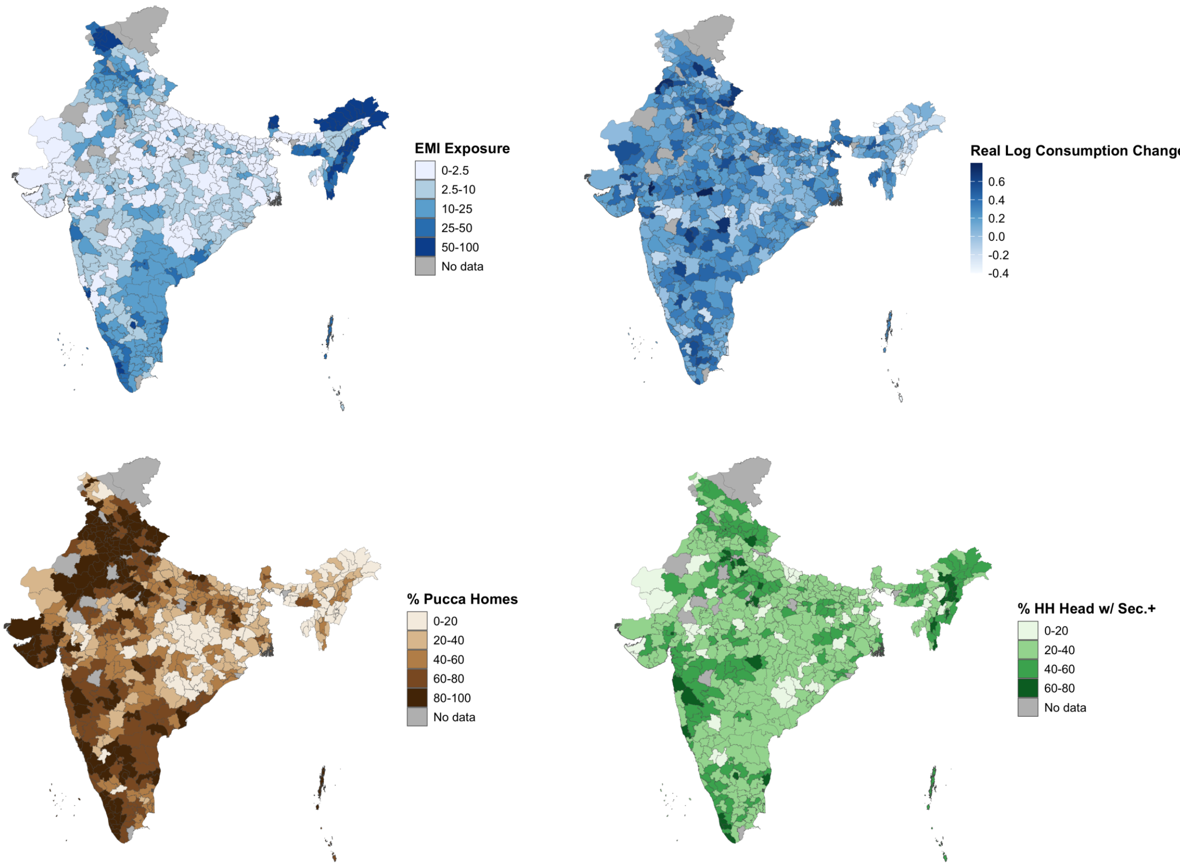

Figure 3: (Clockwise from top left) EMI exposure, consumption growth, pucca (permanent) housing, and household heads with secondary education or more. Data from the 64th round of the NSS 2007-08, "Participation and Expenditure in Education" and "Household Consumer Expenditure."

Table 4: Summary Statistics for 2SLS Model

Variable	Description	Min	1Q	Med	3Q	Max	Mean	SD	N
Treatment and instrument:									
Linguistic distance	Population-weighted mean linguistic distance among mapped speakers with positive distance from Hindi	1.00	2.14	3.13	4.55	5.00	3.22	1.31	573
EMI exposure	Share of children ages 5-19 enrolled in English-medium instruction	0.00	1.42	6.23	17.91	95.98	14.90	20.35	573
Consumption outcomes:									
Real consumption, 2007-08	Person-weighted monthly consumption in common prices	439.46	826.48	986.00	1179.91	2945.06	1035.78	295.72	573
Real consumption, 2017-18	Person-weighted monthly consumption in common prices	561.61	996.59	1174.29	1418.34	4170.07	1261.98	398.55	573
Real log consumption change	Log real consumption in 2017-18 minus log real consumption in 2007-08	-0.86	0.07	0.20	0.34	0.98	0.19	0.22	573
Census 2001 controls:									
Log population	Natural log of Census 2001 district population	10.35	13.61	14.20	14.68	16.08	14.00	1.02	573
Urban population share	Urban population as a percentage of total population	0.00	10.03	18.15	29.73	100.00	23.27	19.11	573
Secondary-plus share, age 7+	Population age 7 and above with matric or higher attainment	4.40	11.39	15.62	21.08	43.67	17.09	7.63	573
Scheduled Caste share	Scheduled Caste population as a percentage of total population	0.00	8.13	15.64	19.98	50.11	14.71	8.64	573
Scheduled Tribe share	Scheduled Tribe population as a percentage of total population	0.00	0.15	3.49	18.27	98.09	16.16	25.85	573
Muslim share	Muslim population as a percentage of total population	0.09	2.59	7.21	13.51	98.49	11.40	14.81	573
Agricultural worker share	Cultivators and agricultural labourers as a percentage of workers	0.00	50.56	66.32	75.37	90.24	60.53	20.40	573
Dependency ratio	Population age 0-14 and 65+ as a percentage of population age 15-64	35.46	58.79	69.09	80.75	100.75	69.39	13.29	573
Electricity access share	Households using electricity for lighting as a percentage of households	3.09	28.36	60.14	78.40	99.71	54.44	28.01	573

Table 4: Summary Statistics for 2SLS Model *(continued)*

Variable	Description	Min	1Q	Med	3Q	Max	Mean	SD	N
Log population density	Natural log of persons per square kilometre	0.88	5.23	5.76	6.41	10.82	5.77	1.16	573

Note: Min. = minimum; 1Q = first quartile; Med. = median; 3Q = third quartile; Max. = maximum; Mean = arithmetic mean; SD = standard deviation; N = number of observations.

Results

The results of our first-stage regression, of EMI exposure on linguistic distance, are provided in the corresponding table in the full report. The results of the second-stage regression, where real log consumption growth is regressed onto the fitted values of EMI exposure, are provided in the corresponding table in the full report. All standard errors in both stages are clustered at the state level.

Discussion

The preferred specification uses person-weighted real log consumption growth as the outcome, the predetermined Census 2001 control set, and Census-2001 state fixed effects. The excluded-instrument partial F is 0.76 with $p = 0.39$. The first-stage coefficient on linguistic distance is 0.51. Conditional relevance is therefore extremely weak.

The second-stage EMI-exposure estimate is 0.06 with $p = 0.453$. Because the instrument contributes almost no conditional first-stage variation, this estimate should not be interpreted as a credible causal effect. The appropriate conclusion is not that EMI has no effect, but that this instrument cannot distinguish such an effect under the preferred control and fixed-effect specification.

The revised design improves measurement and temporal ordering. Consumption is adjusted for state, rural/urban, and survey-month price differences; district means represent people rather than households; controls are measured before treatment; and district histories are harmonized to a Census-2001 geography. Those improvements expose rather than solve the identification problem: most of the raw linguistic-distance relationship with EMI is between states or is shared with predetermined district characteristics.

Future identification work should therefore focus on alternative excluded variation, transparent reduced-form evidence, leave-one-state-out first stages, and weak-IV-robust confidence sets. Nominal log change, real ANCOVA, alternative lineage panels, and the legacy control set remain diagnostic comparisons rather than substitutes for instrument relevance.

Migration and spatial spillovers remain important for interpretation. EMI may raise the welfare of students who later leave their childhood districts, while the district-level outcome records local equilibrium changes rather than individual returns. Remittances, firm location, peer effects, and school-supply responses can therefore move the district estimate away from the individual effect of EMI.

The Census controls reduce concern that the instrument is merely proxying for baseline population, urbanization, education, caste and religious composition, agricultural employment, age structure, electricity access, or density. They do not establish the exclusion restriction, and their joint inclusion with state fixed effects leaves the first stage too weak for conventional 2SLS inference. Spatial diagnostics and lineage-panel sensitivity analyses are reported separately, but neither can restore identifying variation that is absent from the first stage.

District Matching Method

There are three reasons that district labels vary between 2001 to 2007-08 to 2017-18 in the data: genuine political changes, variation in anglicization, and typos. Fuzzy joins (described below) were used to correct for the latter two.

Political changes, however, were not so easily addressed. The vast majority of such changes were one of five kinds: name changes, clean partitions (one district is partitioned into multiple), clean mergers (multiple districts merge into one), carve-outs (*pieces* of multiple districts combine into one), and boundary shifts (India State Stories 2025). Without knowing precisely where in its district each household was at the time of sampling, it is not feasible to match districts across carve-outs and boundary shifts. Our district matching method is thus reliant on the assumption that all district changes were either name changes, clean partitions, or clean mergers.

This seems to be precisely the same assumption made by India State Stories (2025) and Jaacks et al. (2020), whose data, despite containing some factual errors and typos, was the basis of our district matching method.

Table 5

First-Stage Regression: EMI Exposure on Linguistic Distance

	EMI Exposure
	(1)
Linguistic distance	0.514
	(0.592)
Log population	0.663
	(0.694)
Urban population share	0.116*
	(0.051)
Secondary-plus share, age 7+	0.570***
	(0.168)
Scheduled Caste share	0.030
	(0.069)
Scheduled Tribe share	0.047
	(0.030)
Muslim share	0.060
	(0.044)
Agricultural worker share	-0.045
	(0.053)
Dependency ratio	-0.058
	(0.066)
Electricity access share	-0.050
	(0.043)
Log population density	0.983
	(0.855)
Intercept	30.458***
	(8.512)
Observations	573
R^2	0.894
Adjusted R^2	0.885
Residual Std. Error	6.899
Model's F-Statistic	98.93
Instrument's F-Statistic	0.76

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Standard errors clustered by state in parentheses.

Table 6

Second-Stage Regression: Real Log Consumption Growth on EMI Exposure (Fitted)

	Real Log Consumption Growth
	(1)
EMI exposure (fitted)	0.064
	(0.086)
Log population	-0.028
	(0.069)
Urban population share	-0.003
	(0.010)
Secondary-plus share, age 7+	-0.035
	(0.051)
Scheduled Caste share	-0.004
	(0.006)
Scheduled Tribe share	-0.005
	(0.004)
Muslim share	-0.003
	(0.005)
Agricultural worker share	0.004
	(0.005)
Dependency ratio	0.010
	(0.007)
Electricity access share	0.003
	(0.005)
Log population density	-0.086
	(0.109)
Intercept	-2.662
	(2.929)
Observations	573
R^2	-3.545
Adjusted R^2	-3.934
Residual Std. Error	0.498
F-Statistic	

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Standard errors clustered by state in parentheses.

To justify this assumption we turn to Kumar and Somanathan (2016), who up until 2001 are able to track the *proportion* of each district's population that was allocated into a new district as a result of district changes. They label these changes as one of two types: clean partitions and non-partitions (which would include name changes, clean mergers, carve-outs, and border shifts). By extracting their results using Tabula (Aristarán et al. 2018), we know that 93.4% of non-partitions were either name changes or transfers of uninhabited land (Kumar and Somanathan 2016). These district changes are plotted in the corresponding figure in the full report.

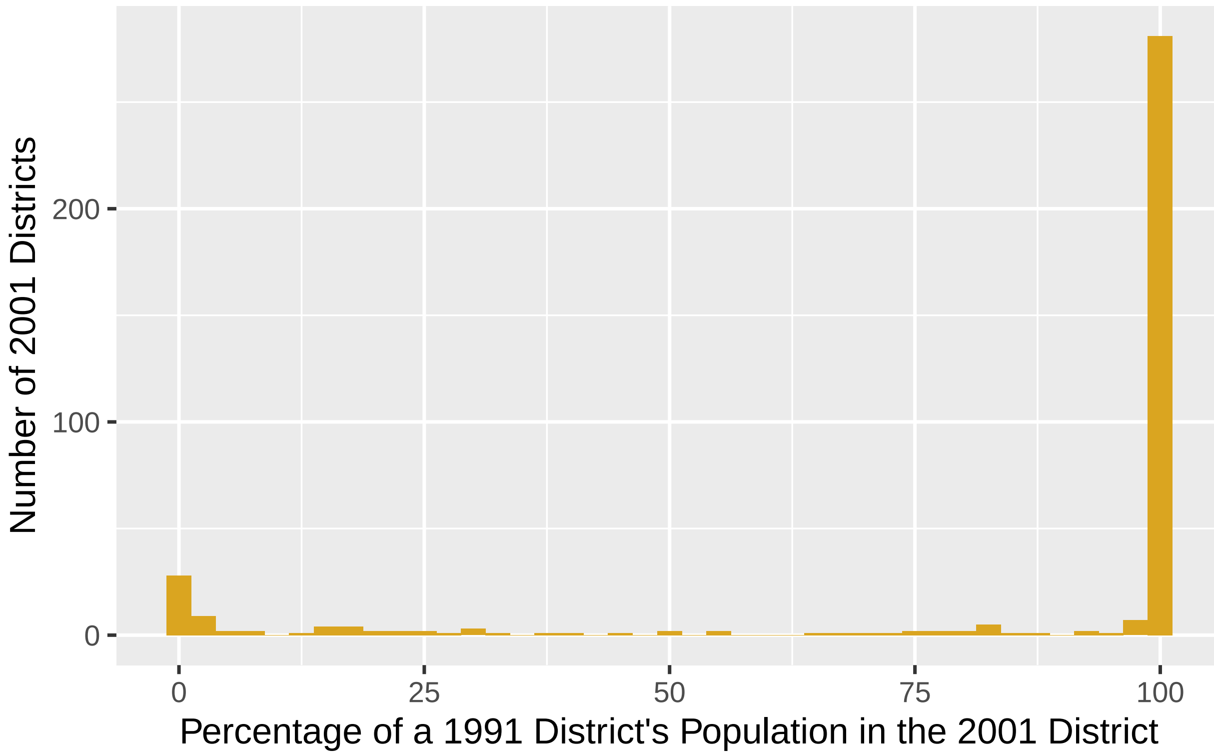

Figure 4: Number of 2001 districts which absorbed a percentage of a 1991 district's population via name change, clean merger, carve-out, or border shift. Data from Kumar & Somanathan (2016).

We thus assume no district changes were carve-outs or border shifts. To match districts, we ran a hierarchical, cascading sequence of iterated fuzzy joins. Data from 2001, 2007-08, and 2017-18 is first linked to the closest year in the district trackers of India State Stories (2025) and Jaacks et al. (2020). Districts are then matched if they meet a certain string/phonetic distance threshold. Matches were made in the following order:

1. `soundex` $= 0$, for phonetically equivalent variations in anglicization (e.g., "Baleswar" and "Balasore").
2. `qgram` distance $= 0$, for rearrangements of bigrams (e.g., "East Godavari" and "Godavari East").
3. Jaro-Winkler distance ≤ 0.15, for respellings and vowel swaps with 0.85 degrees of similarity.
4. Full Damerau-Levenshtein distance ≤ 2, for ≤ 2 insertions, deletions, substitutions, and transpositions.
5. Optimal String Alignment distance ≤ 1, for a single typo.

Any district from the 2001, 2007-08, or 2017-18 data which remains unmatched after this process is then matched with the next closest year in the district tracker. This process repeats until all years in the tracker had been exhausted. 80 still-unmatched districts were then manually matched.

The active harmonization now uses Census-2001 districts as the common unit. Counts are summed, means and shares are recomputed from pooled numerators and denominators, and Gini coefficients are reconstructed from pooled household records where support permits. Conservative and broader reviewed lineage panels are retained as sensitivity analyses. State fixed effects are now estimable, but the excluded instrument has almost no conditional relevance once those fixed effects and the Census controls are included.

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